

EE/CE 252/251

Signals and Systems

with
Dr. Naveed R. Butt
@
Habib University

Vocabulary

- زبانِ یارِ مَن تُرکی



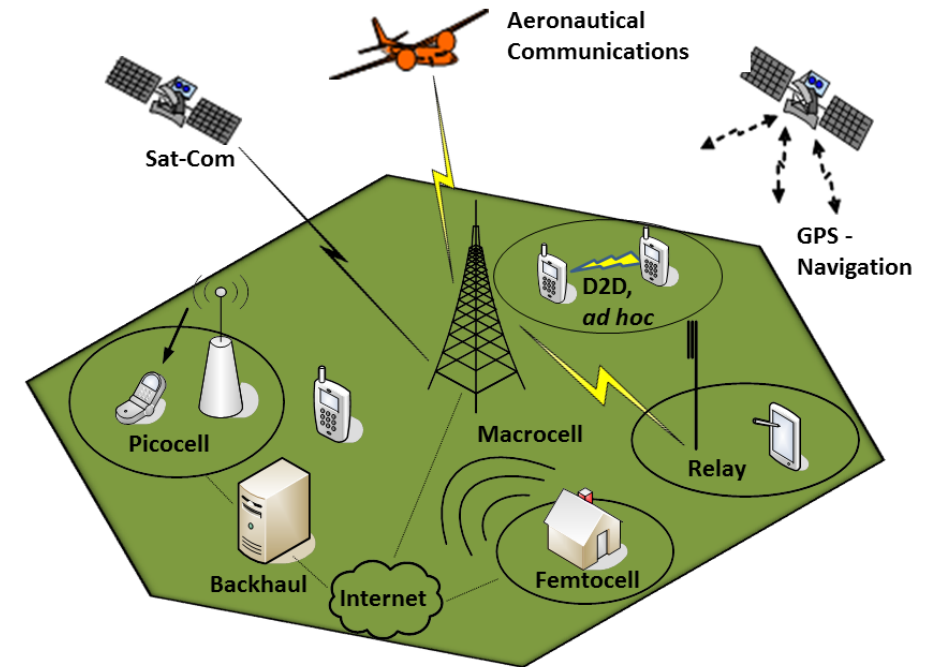
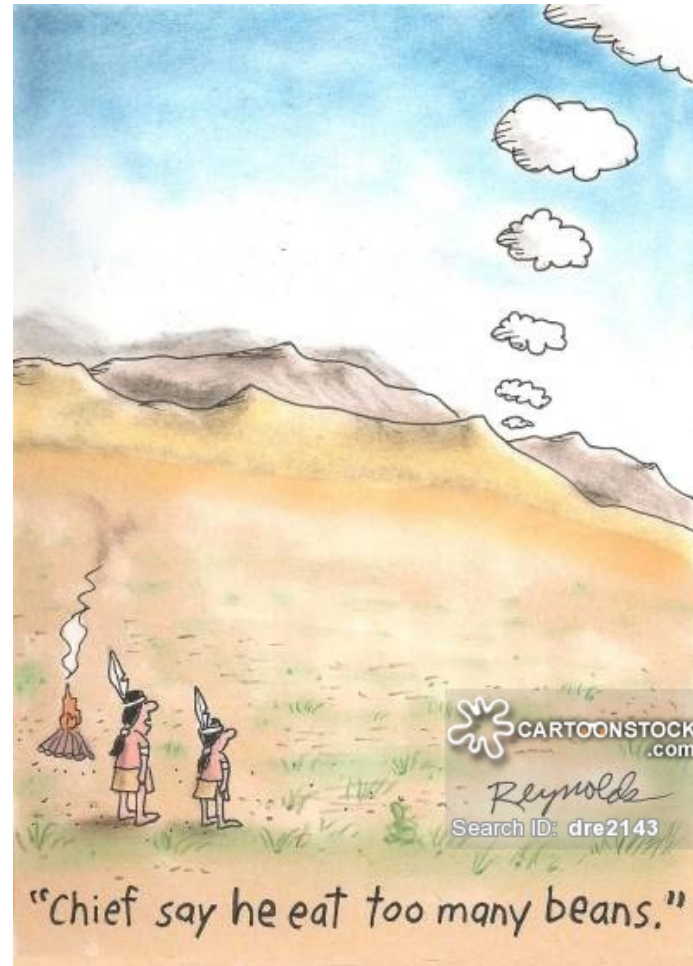
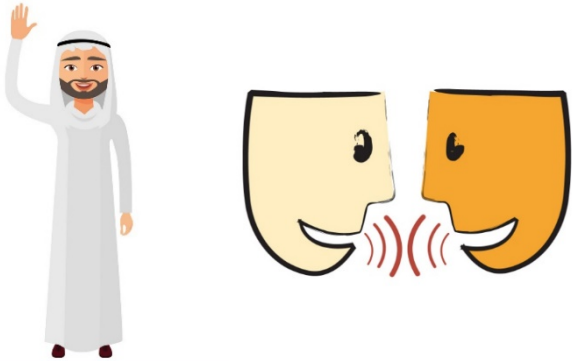
What a moose hears

Vocabulary – Signals & Systems

- Signal
- System
- Continuous-Time
- Discrete-Time
- Impulse
- Step
- Transform
 - Time Domain
 - Frequency Domain
- Response
- Convolution

*What is a
signal?*

Signal



Signal

A set of data or information!

Signal

How do we normally represent data/information?

Signal

How do we normally represent data/information?

Tabular

.9998
.9994
.9986
.9976
.9962

.9945
.9925
.9903
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.8572
.8480
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.8290
.8192

.8090

Signal

How do we normally represent data/information?

Model/Function

$$\cos(t)$$

Tabular

.9998
.9994
.9986
.9976
.9962

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.9925
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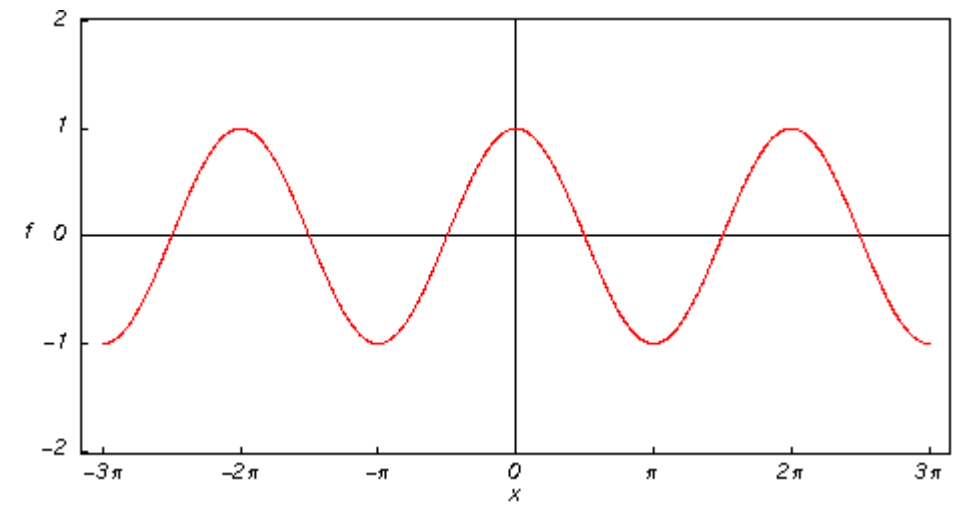
.8572
.8480
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.8099

Model/Function

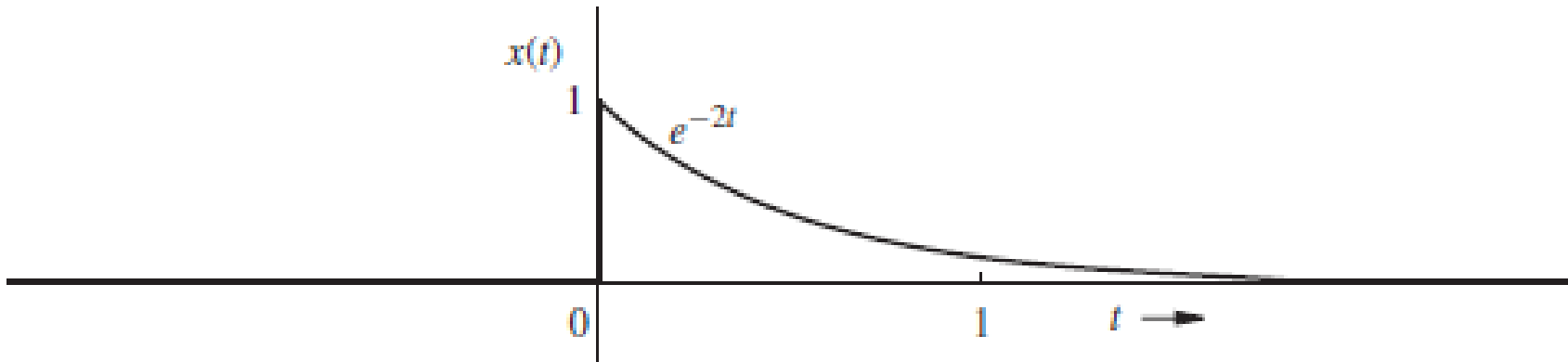
$$\cos(t)$$

Graph



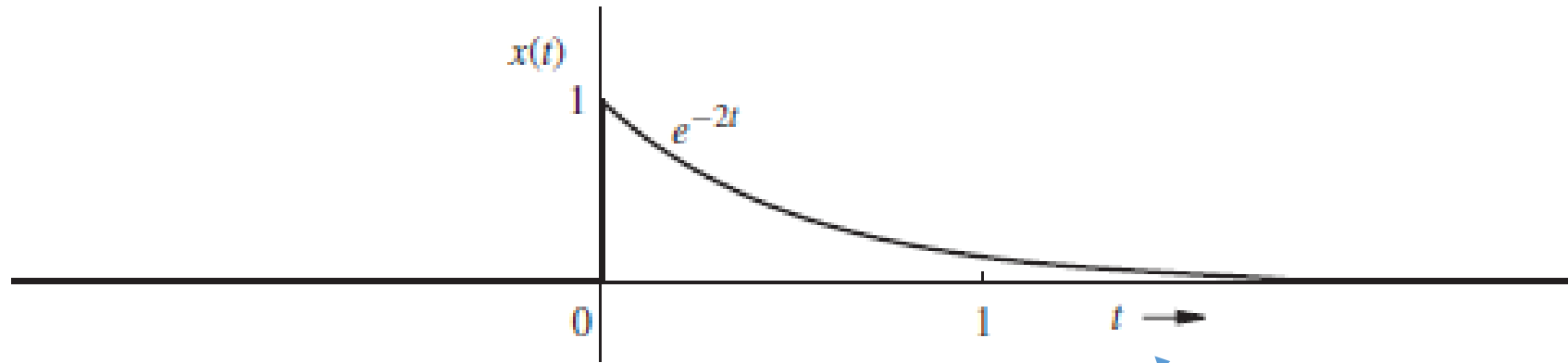
Signal

Mathematically and Graphically speaking...



Signal

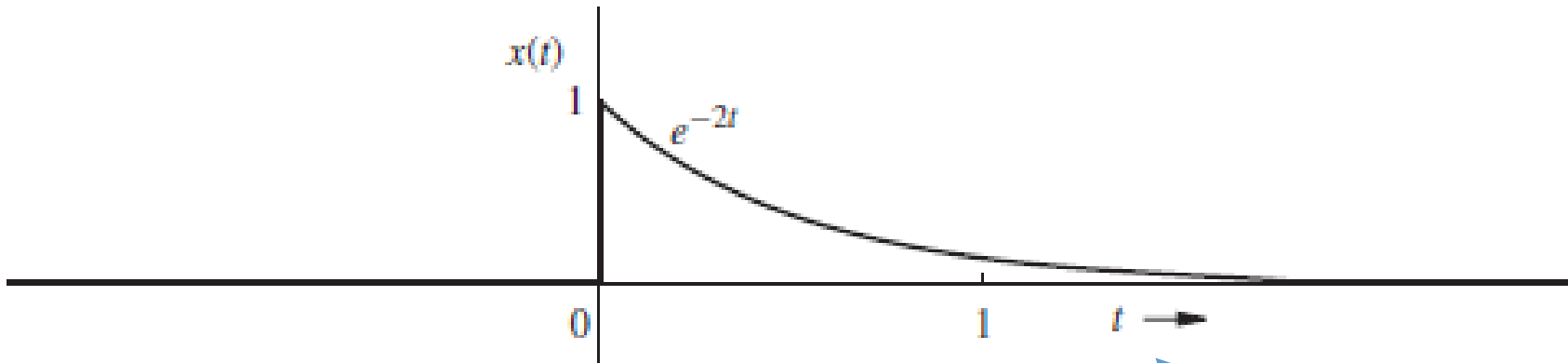
Mathematically and Graphically speaking...



Our x-axis is often
“time” (but doesn’t
have to be).

Signal

Mathematically and Graphically speaking...

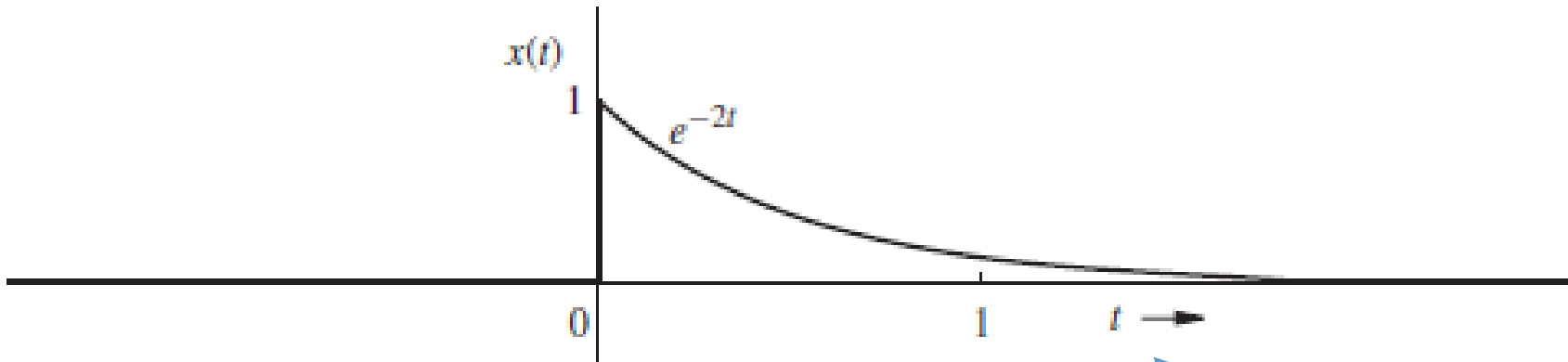


Can you write the above signal as a function?

Our x-axis is often "time" (but doesn't have to be).

Signal

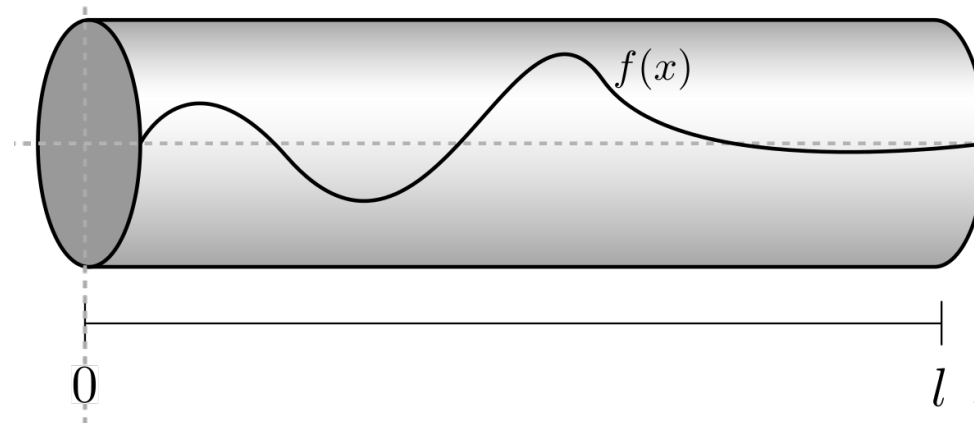
Mathematically and Graphically speaking...



$$x(t) = \begin{cases} e^{-2t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Our x-axis is often "time" (but doesn't have to be).

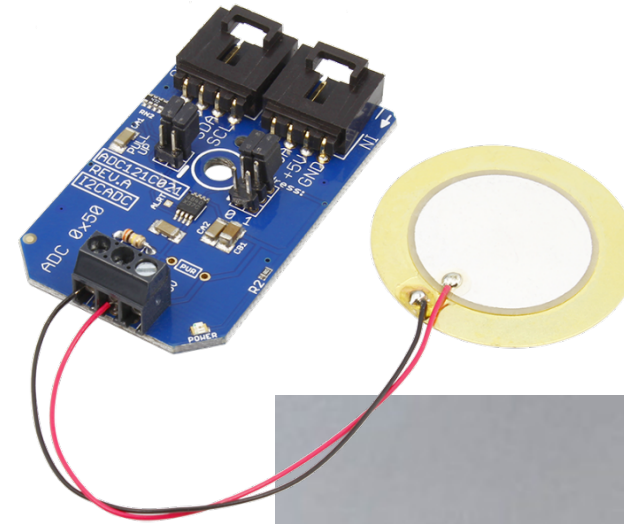
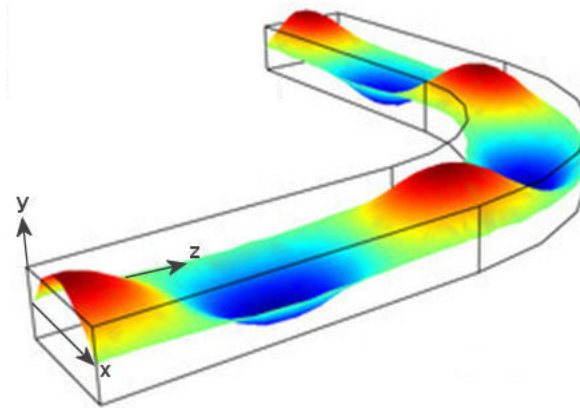
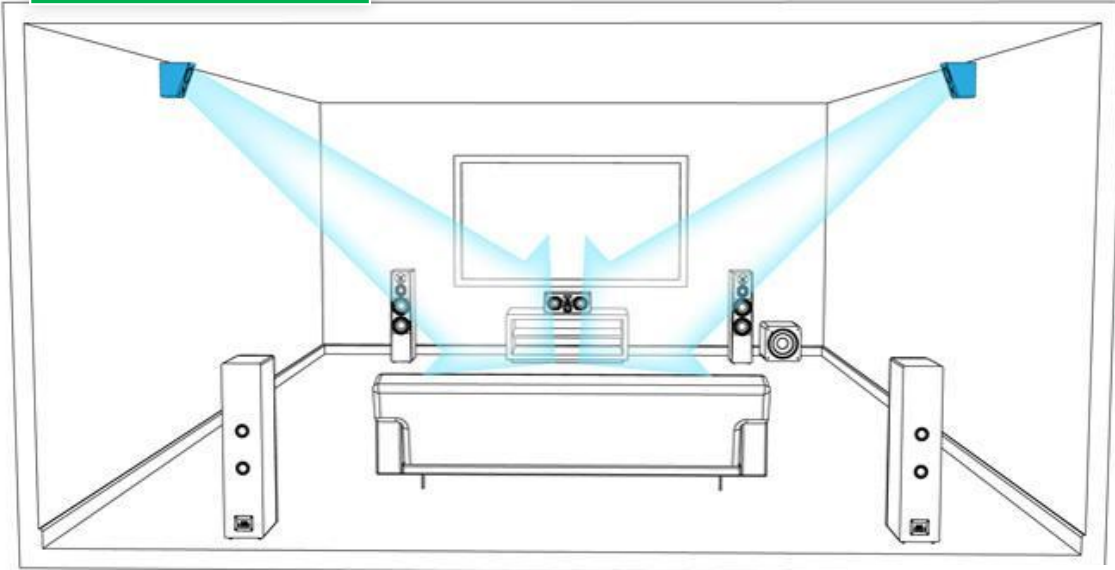
Signal



For temperature across a rod, x-axis could be along the length.

*What is a
system?*

System





An entity that processes a signal

**Syste
m**

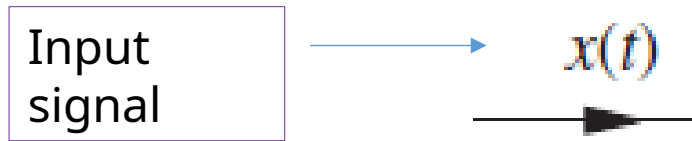
Including “generates” or
“receives”

An entity that processes a signal

System

Including “generates” or
“receives”

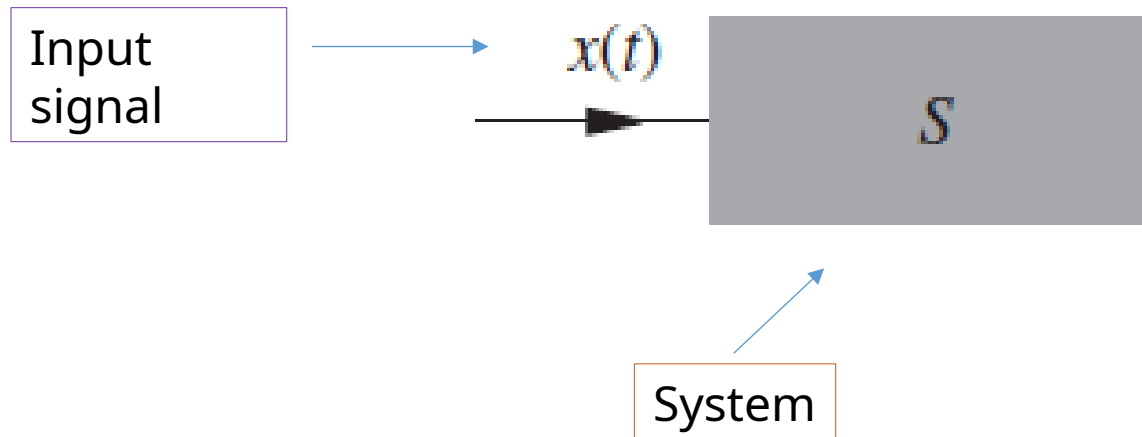
An entity that processes a signal



System

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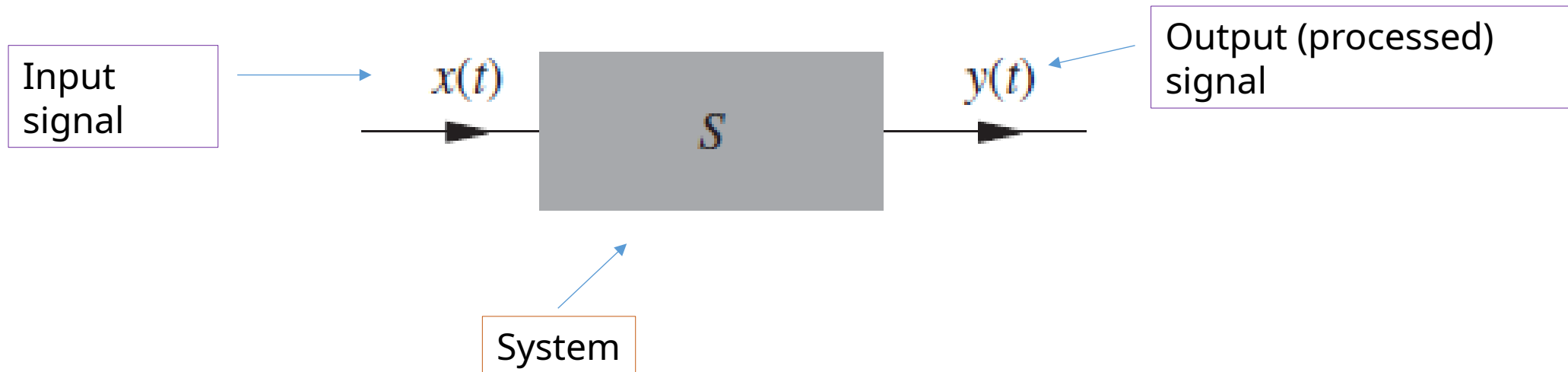
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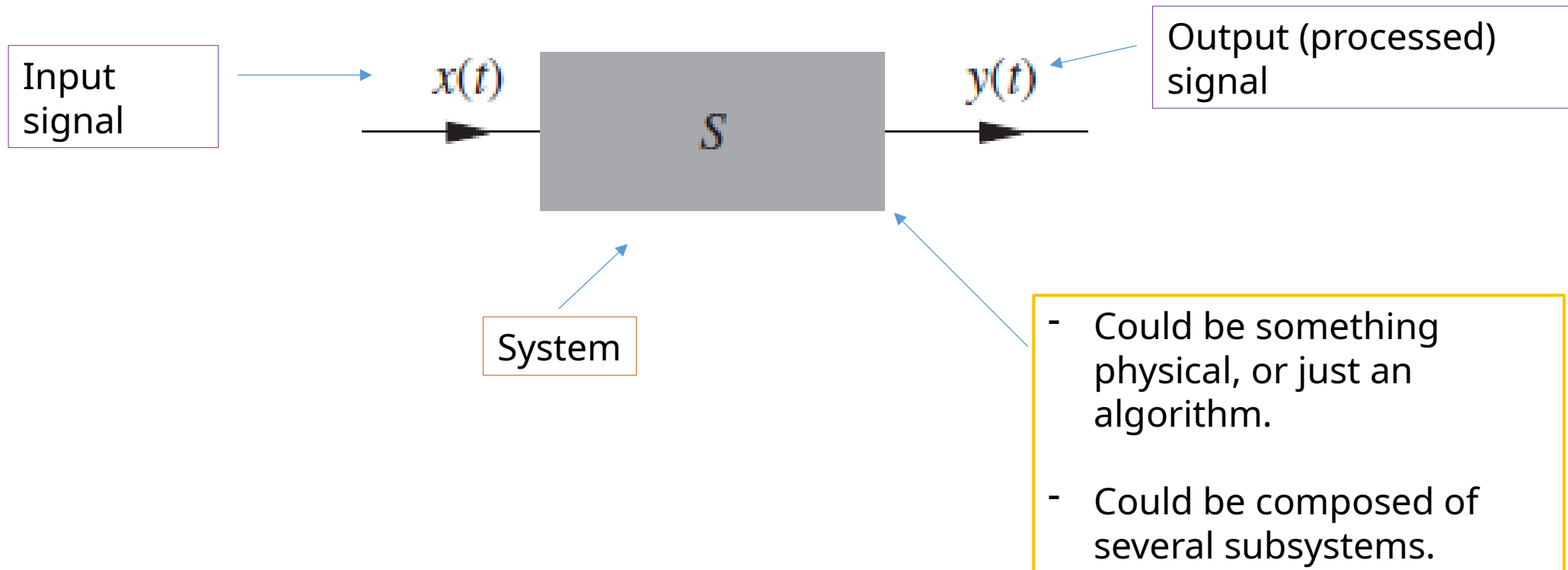
An entity that processes a signal



System

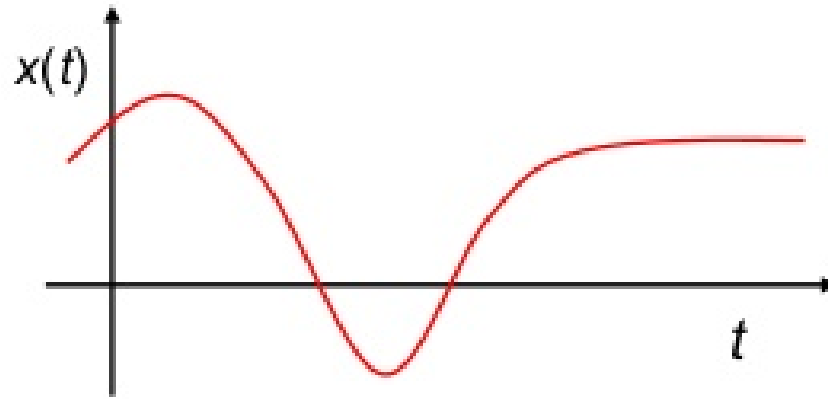
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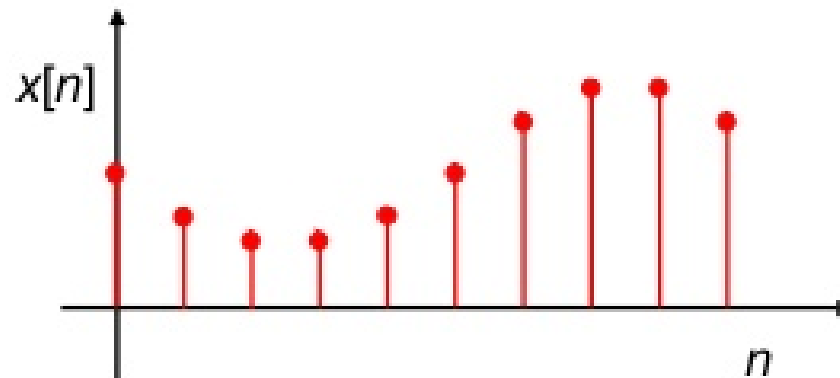


Continuous vs. Discrete

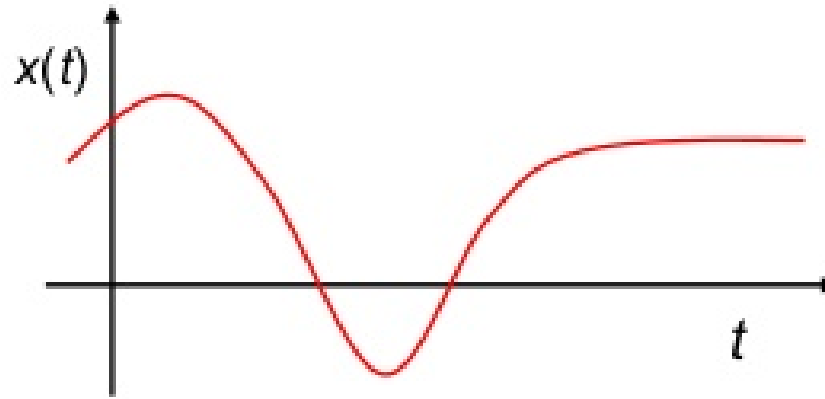
“Continuous-Time”



“Discrete-Time”

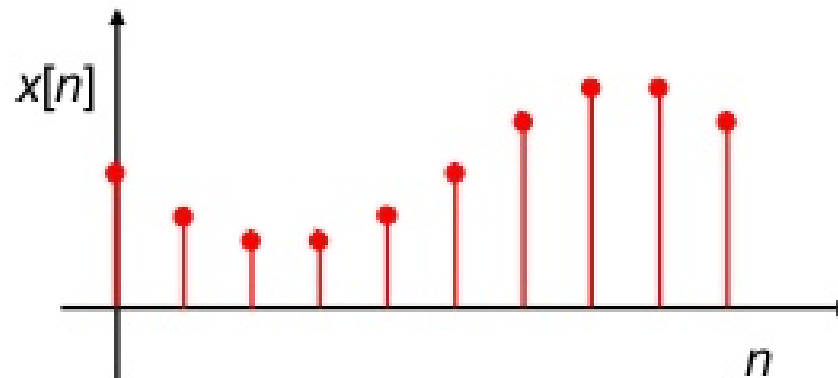


“Continuous-Time”

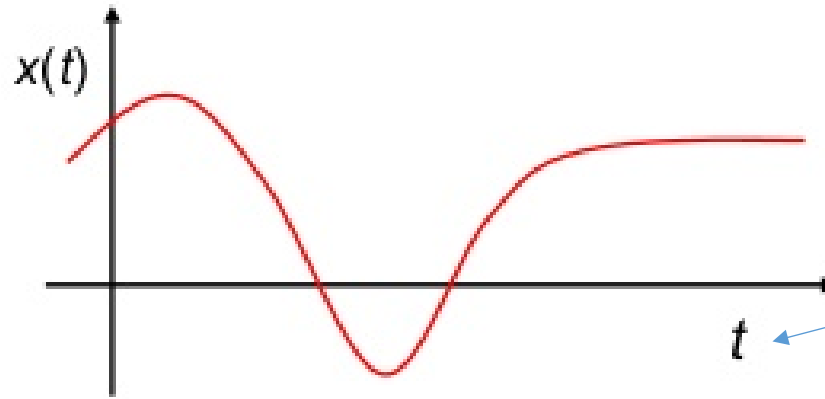


Q. What's the difference?

“Discrete-Time”

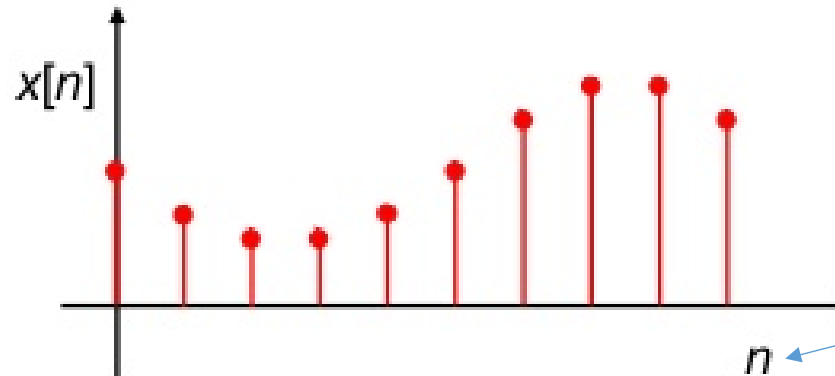


“Continuous-Time”



X-axis is a continuous variable (can take any value in a given range).

“Discrete-Time”



X-axis is a discrete variable (cannot take all values in a given range, e.g. $n = 1, 2, 3, \dots$)

**Impuls
e**

**Impuls
e**

(ever heard of “impulsive”
behavior?)

Impuls e

impulse

noun

UK  /'ɪm.pʌls/ US  /'ɪm.pʌls/

impulse *noun* (WISH)



C2 [C + to infinitive]

a sudden strong wish to do something:



Impuls e

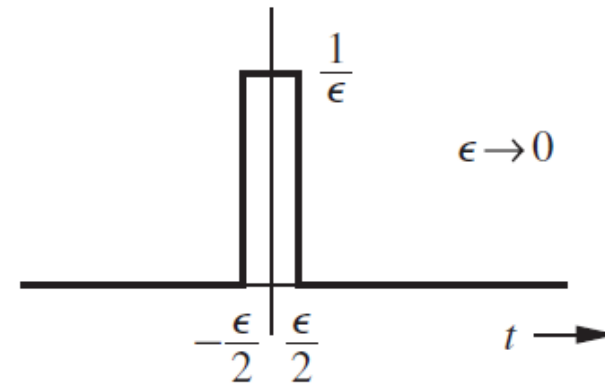


Sometimes we want to give a very **brief** “touch” to a system to see its reaction.

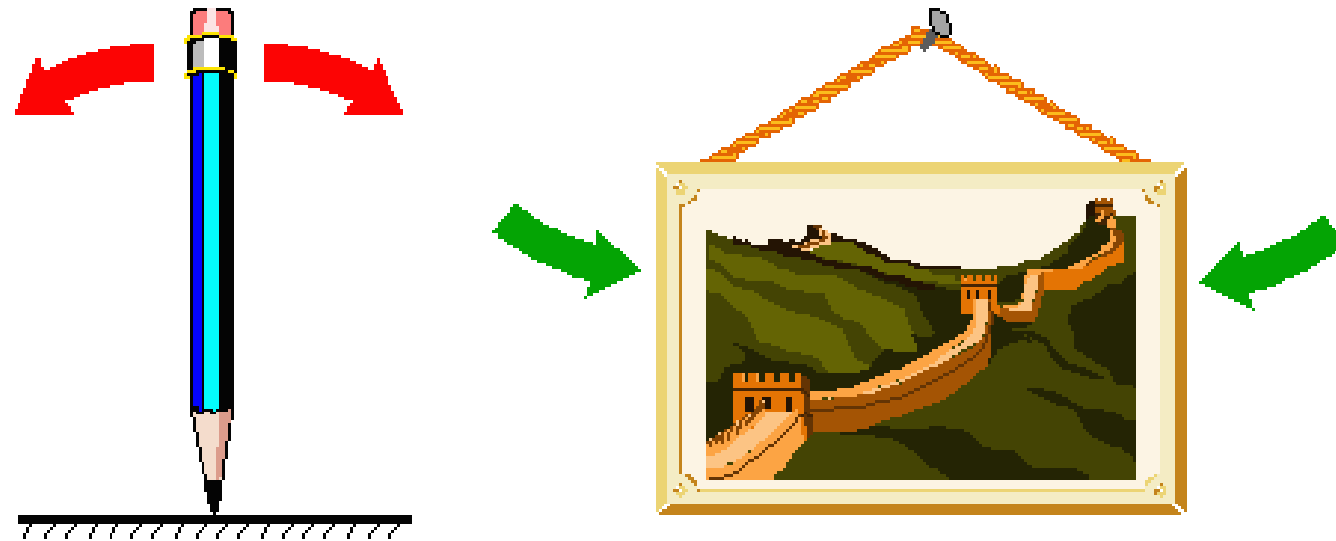
Impulse



Sometimes we want to give a very **brief** “touch” to a system to see its reaction.



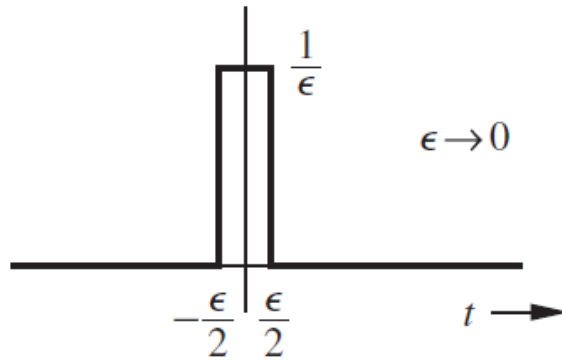
We could use something as simple as an impulse to study response of a system...



Impulse

Mathematically and graphically speaking...

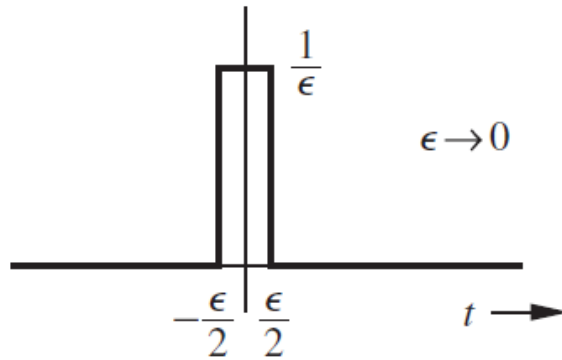
Continuous-Time Case



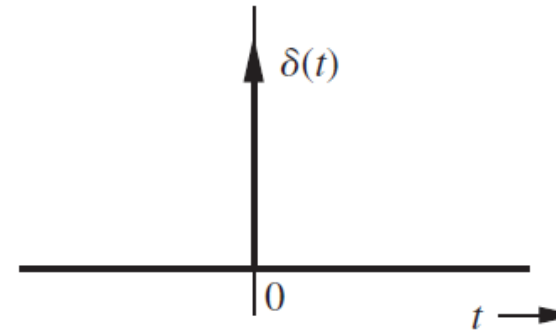
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



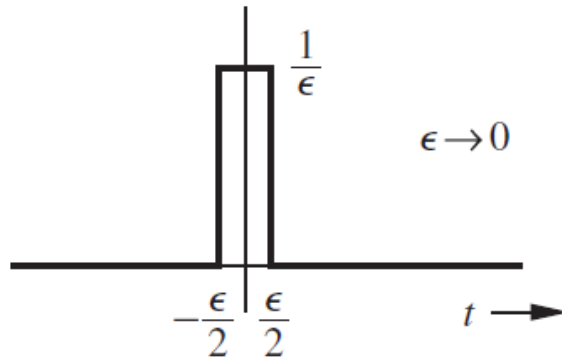
We normally draw it as ...



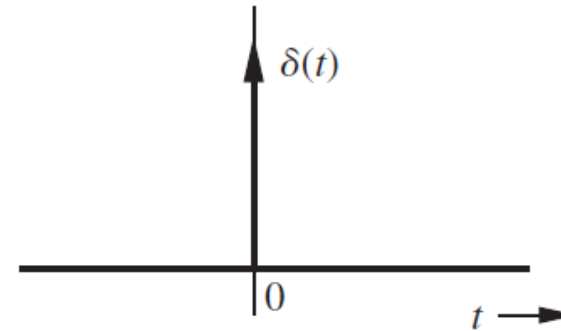
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



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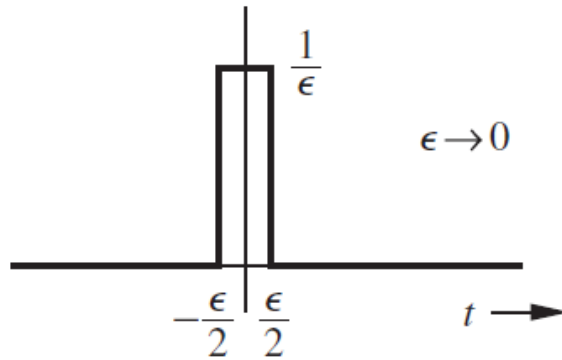


Two conditions that define a CT Impulse

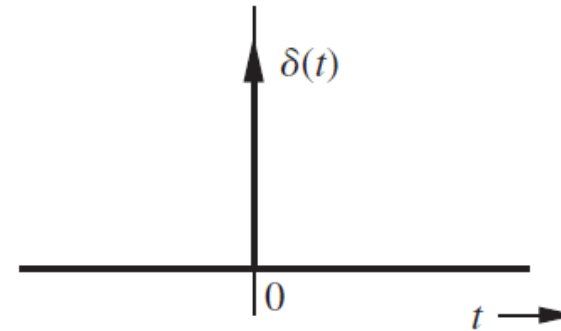
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



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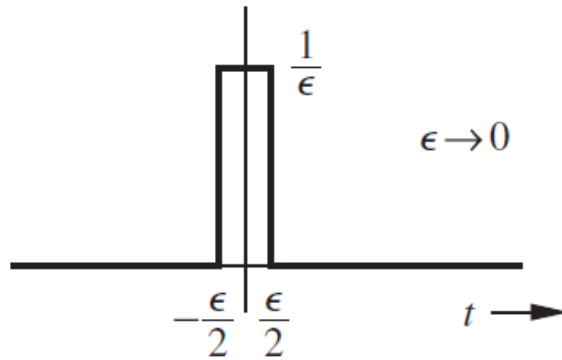
$$\delta(t) = 0 \quad t \neq 0$$

Two conditions that define a CT Impulse

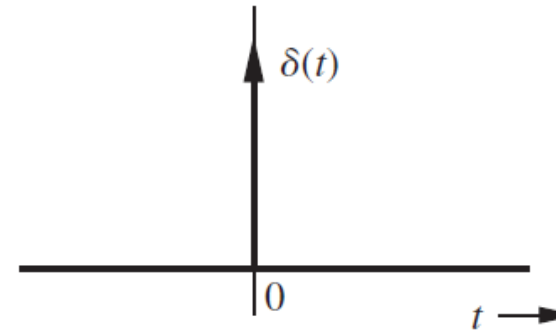
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



We normally draw it as ...



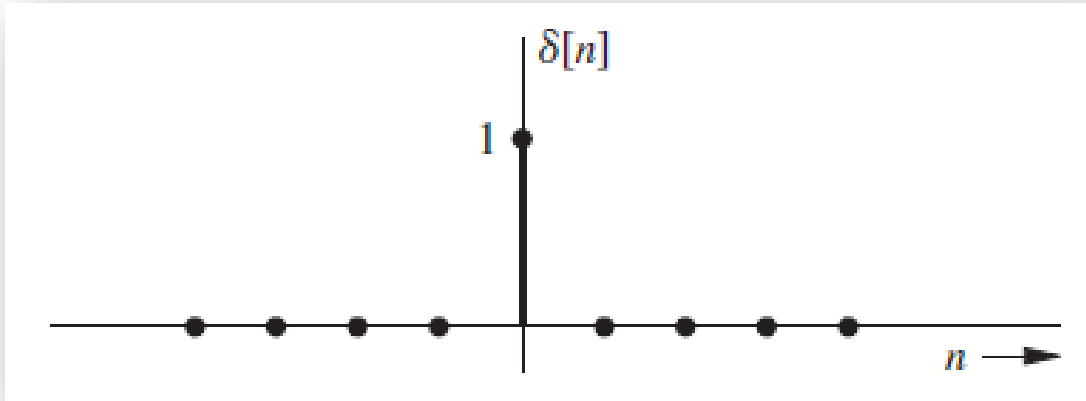
$$\delta(t) = 0 \quad t \neq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

Two conditions that define a CT Impulse

Impulse

Mathematically and graphically speaking...

Discrete-Time Case

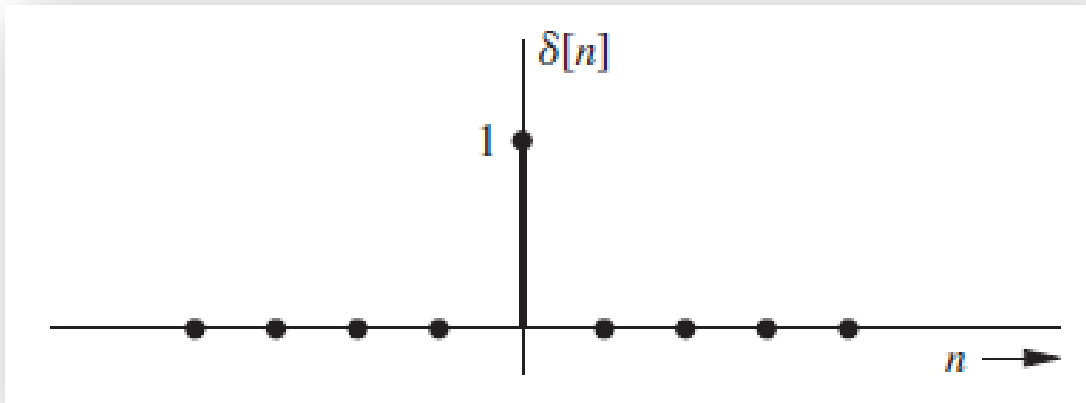


Can you write this as a function?

Impulse

Mathematically and graphically speaking...

Discrete-Time Case



$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

Step

Sometimes we want to give a **sustained** “push” to a system to see its reaction (or what happens).

Step

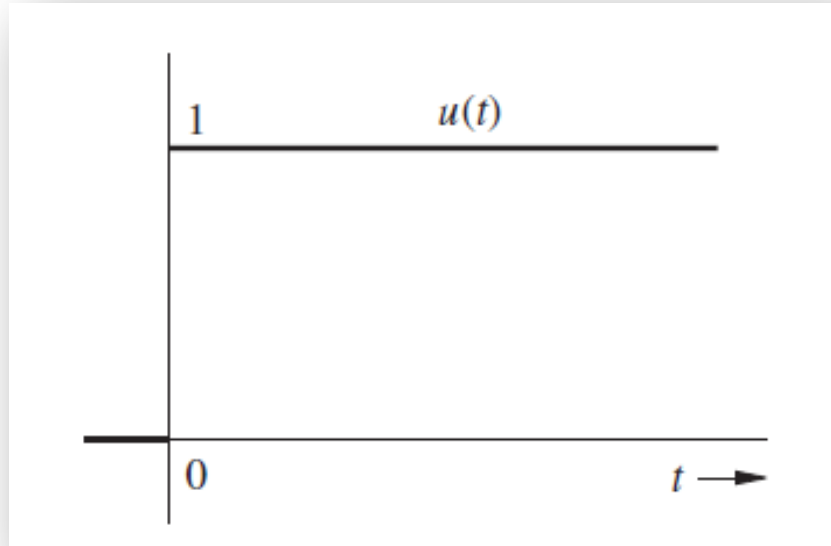
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Mathematically and graphically speaking...

Continuous-Time Case

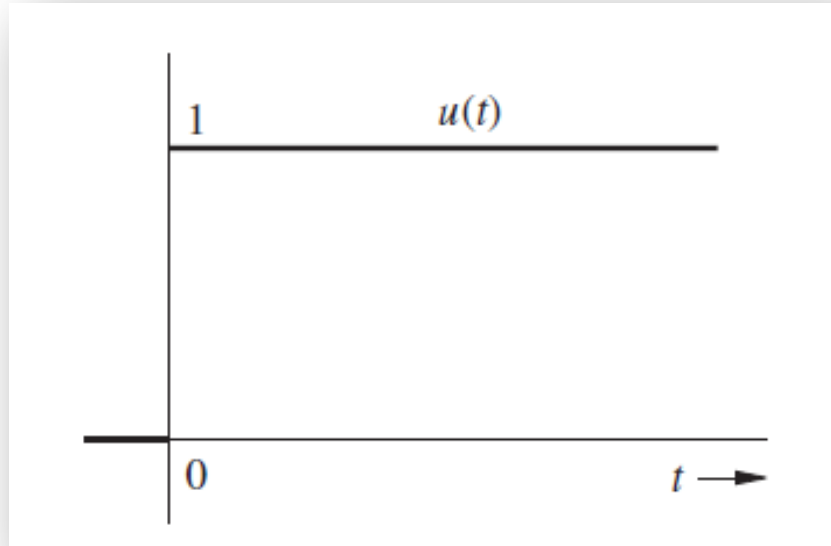


Can you write this as a function?

Step

Mathematically and graphically speaking...

Continuous-Time Case

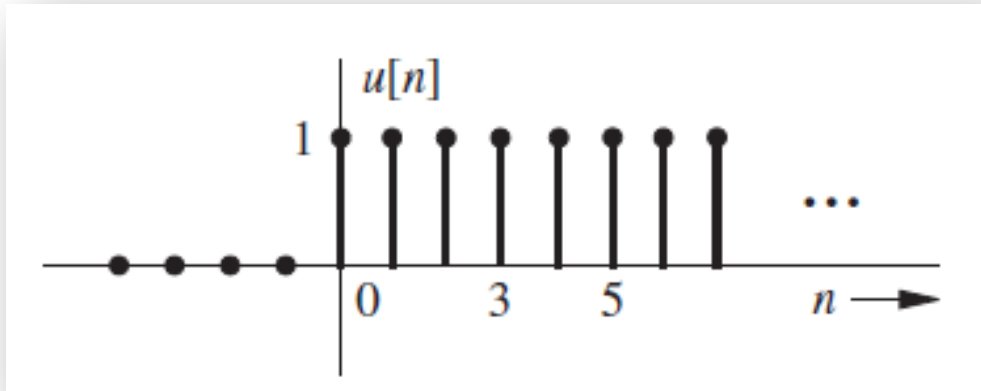


$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Step

Mathematically and graphically speaking...

Discrete-Time Case

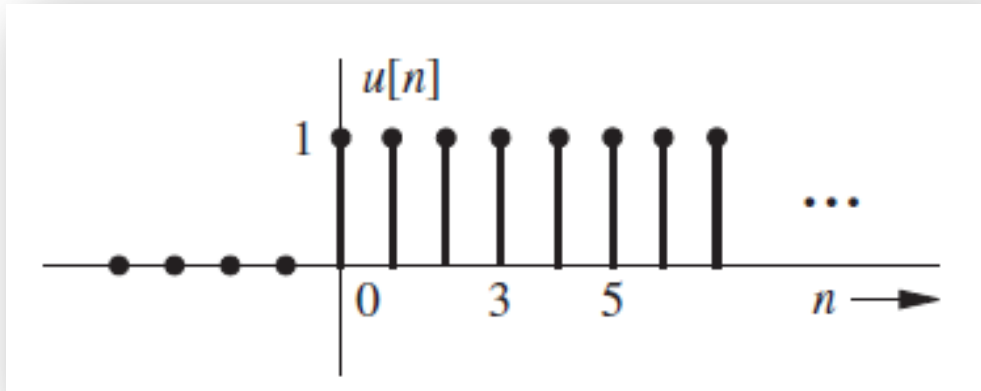


Can you write this as a function?

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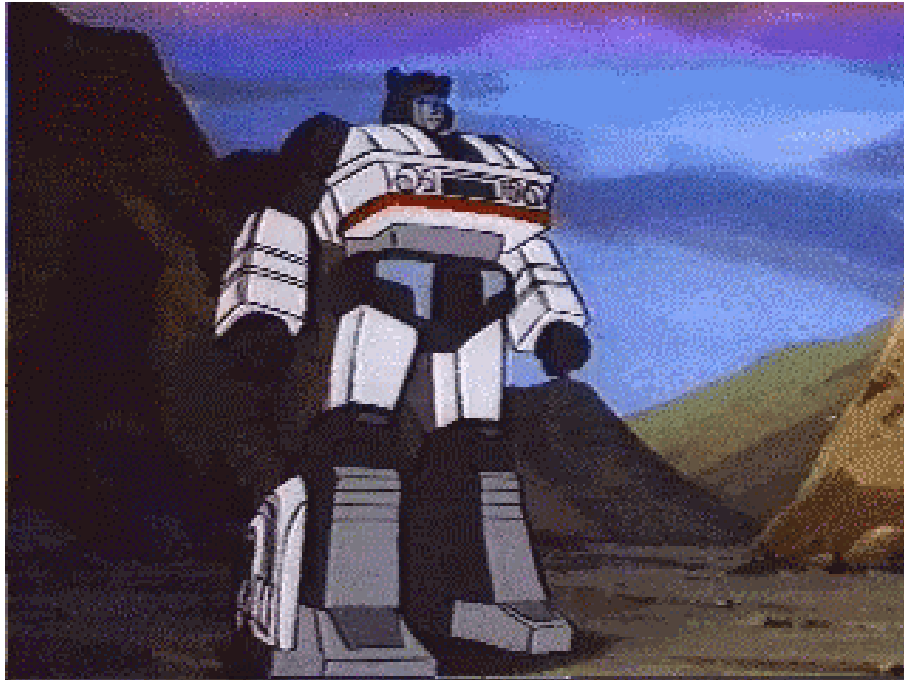
Mathematically and graphically speaking...

Discrete-Time Case



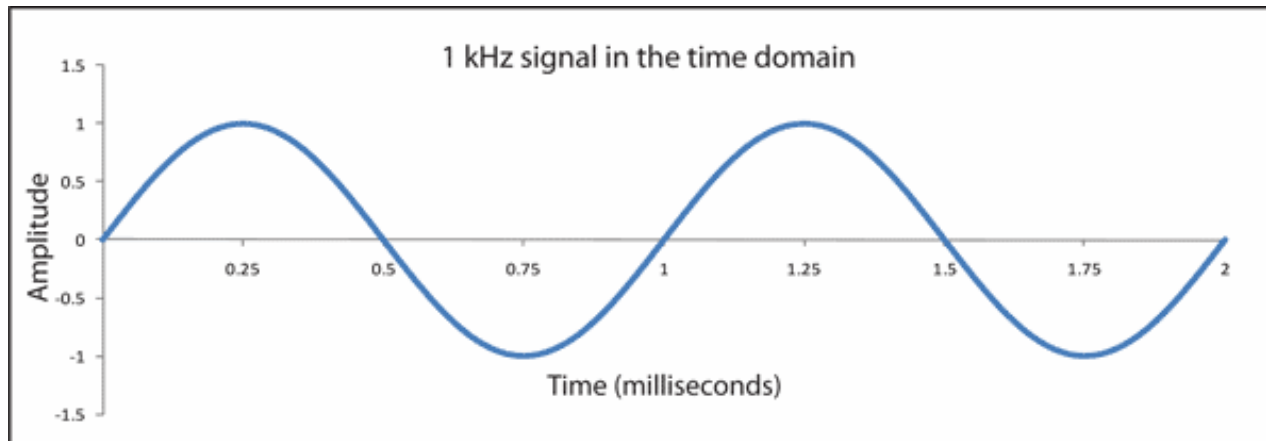
$$u[n] = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

Transform



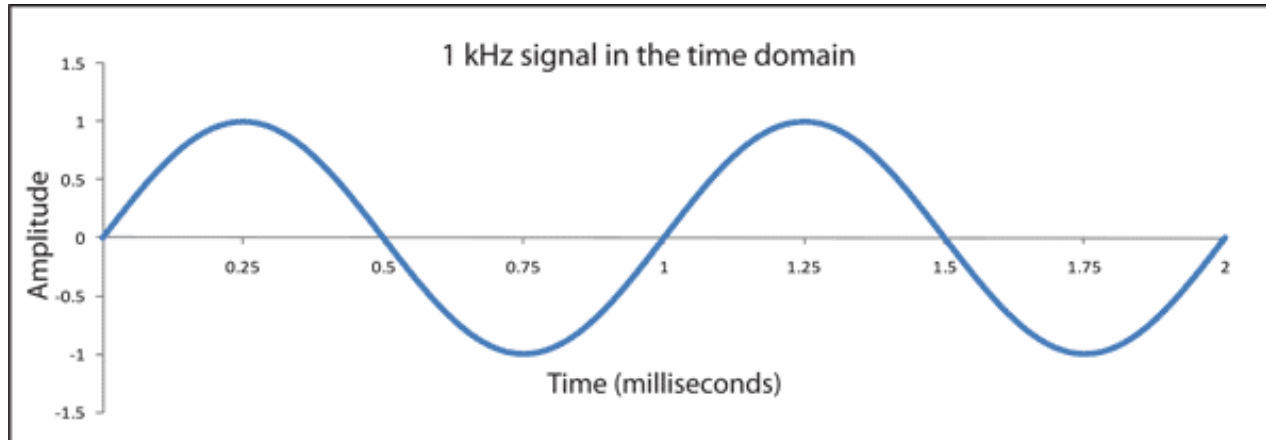
A transform is an **alternate form or representation** of something.

Transform



This is clearly a sine function.
Only information I need to
record is:

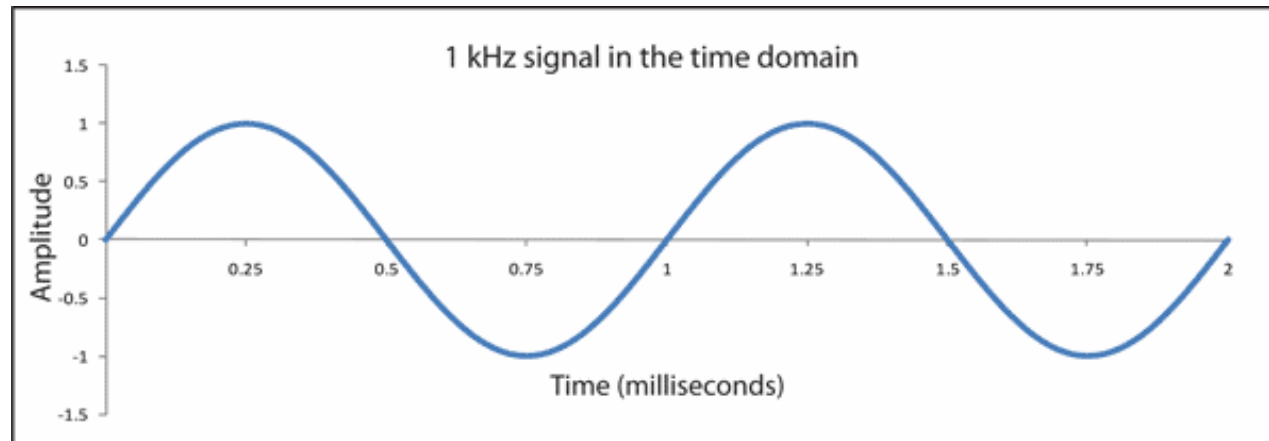
Transform



This is clearly a sine function.
Only information I need to record is:

- (a) This is a sine function
- (b) It has amplitude 1
- (c) It has frequency 1 kHz

Transfor m



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(later we will bring in
phase information as
well)

Transfor m

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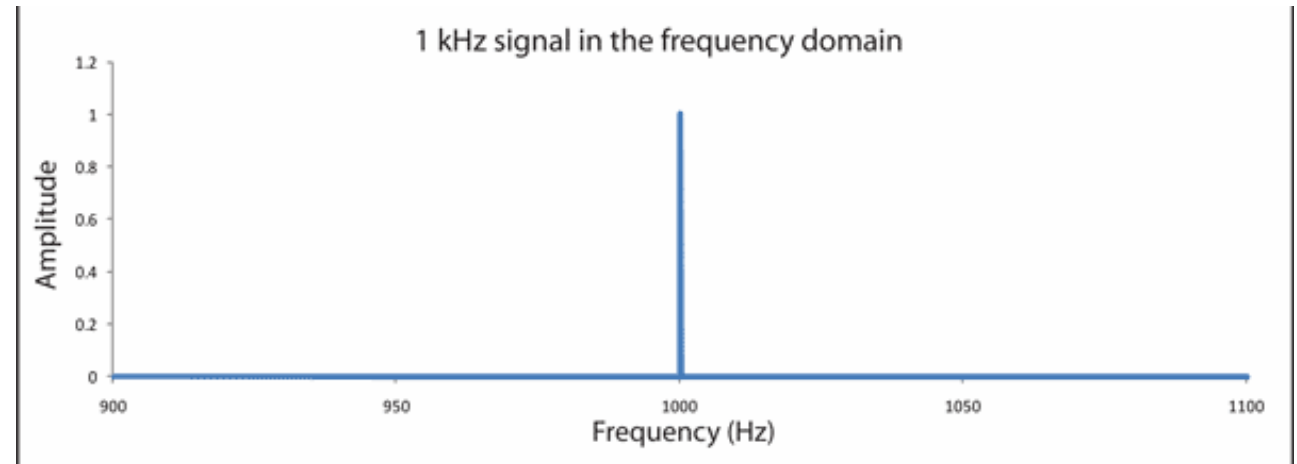
Let's plot this information on a graph
whose x -axis is frequency and y -axis
is amplitude of sine functions...

Transform

This is clearly a sine function.
Only information I need to record is:

- (a) This is a sine function
- (b) It has amplitude 1
- (c) It has frequency 1 kHz

Let's plot this information on a graph
whose x -axis is frequency and y -axis
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Sine functions represented by
their amplitude and frequency
only...

Transform

A transform is an **alternate form or representation** of something.

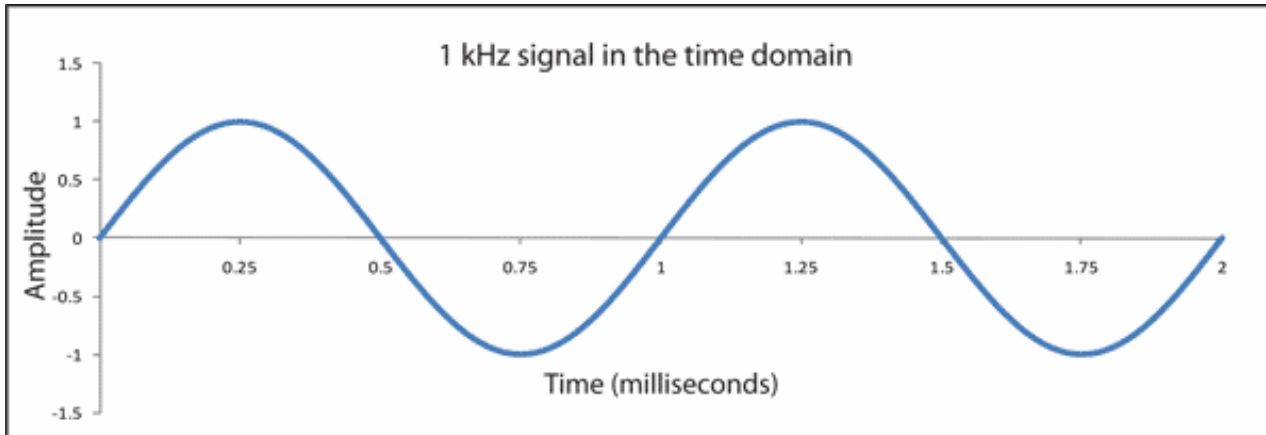
Transform

A transform is an **alternate form or representation** of something.

Two alternate representations
of $\sin(1000t)$

Transform

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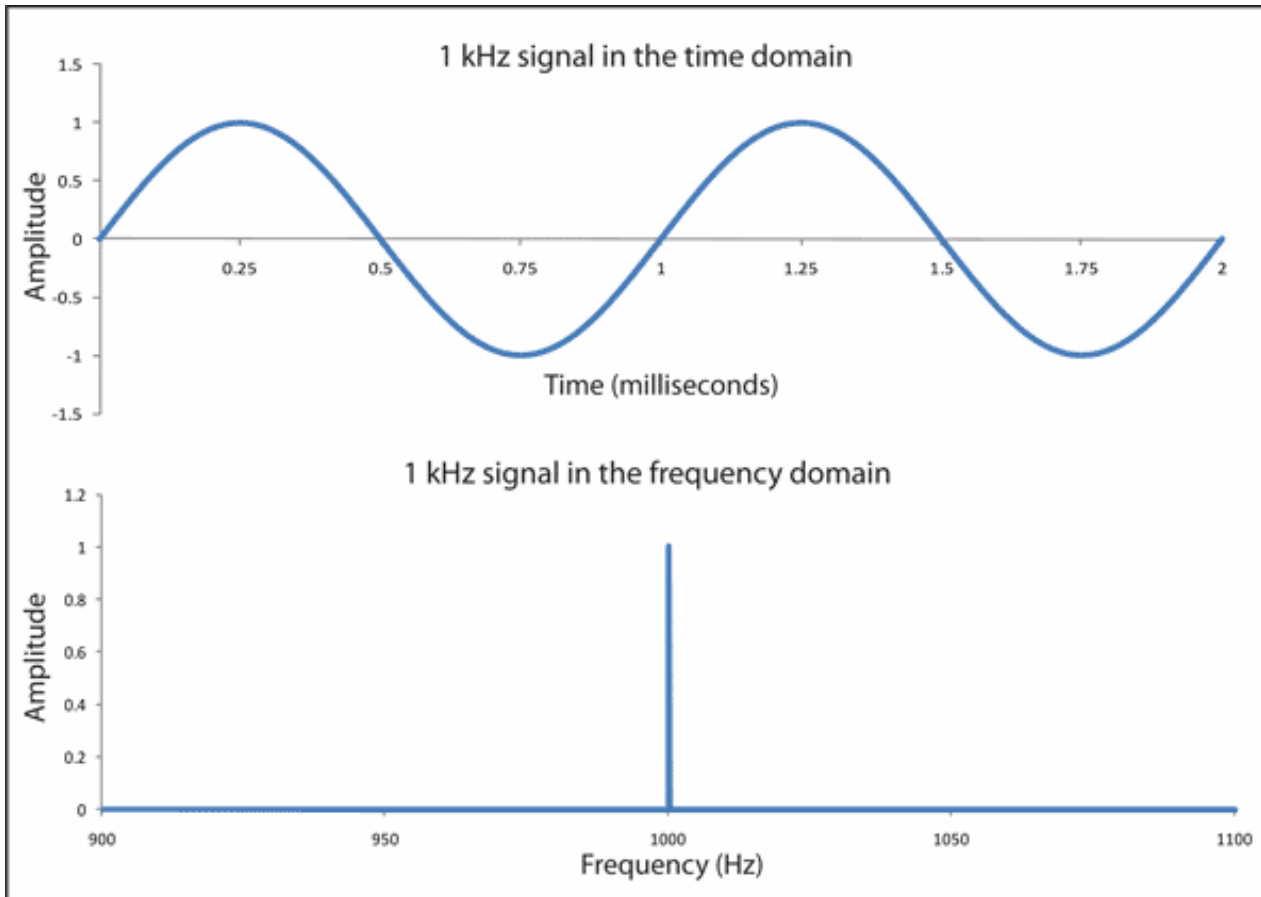


Time Domain (amplitude vs. time)

Two alternate representations
of $\sin(1000t)$

Transform

A transform is an **alternate form or representation** of something.



Time Domain (amplitude vs. time)

Two alternate representations
of $\sin(1000t)$

Frequency Domain (amplitude vs. freq.)

Response

Response

Response is what a system does to a signal entering it.

Response

Response is what a system does to a signal entering it.



Response

Response is what a system does to a signal entering it.

A corrupt system...

The Case of Double Shah



Response

Response is what a system does to a signal entering it.

A corrupt system...

The Case of Double Shah



Give me one rupee, and I'll
give you two!

Response

Response is what a system does to a signal entering it.

A corrupt system...

The Case of Double Shah



Give me one rupee, and I'll
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Response

Response is what a system does to a signal entering it.

A corrupt system...

The Case of Double Shah



Give me one rupee, and I'll give you two!

System "response", let's call it h , then $y = hx$ with $h = 2$.

Convolutio n

(ever heard of a convoluted argument?)

Convolutio n

(ever heard of a convoluted argument?)



convoluted

/ˌkɒnvəˈl(j)uːtɪd/

adjective

1. (especially of an argument, story, or sentence) extremely complex and difficult to follow.
"the film is let down by a convoluted plot in which nothing really happens"

Similar:

complicated

complex

involved

intricate

elaborate

impenetrable



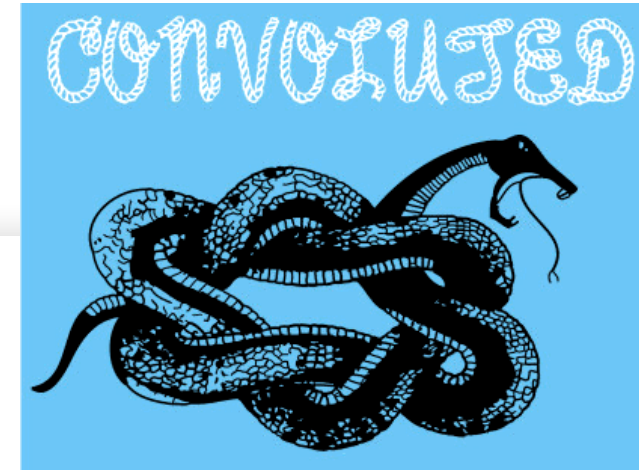
2. **TECHNICAL**

intricately folded, twisted, or coiled.

"walnuts come in hard and convoluted shells"

Convolutio n

(ever heard of a convoluted argument?)



convoluted

/ˌkɒnvəˈl(j)uːtɪd/

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intricately folded, twisted, or coiled.

"walnuts come in hard and convoluted shells"

Convolutio n

Mathematically speaking...

Convolutio n

Mathematically speaking...

**A MATHEMATICAL OPERATION
(JUST AS $+$, $-$, $\times$, $\div$) THAT HELPS US
CALCULATE THE RESPONSE OF A
SPECIAL TYPE OF SYSTEMS.**

Convolutio n

Mathematically speaking...

**A MATHEMATICAL OPERATION
(JUST AS $+$, $-$, $\times$, $\div$) THAT HELPS US
CALCULATE THE RESPONSE OF A
SPECIAL TYPE OF SYSTEMS.**

Convolution is often denoted by $*$

The Curious (and Completely Made Up) Case of Quadruple Shah

The Curious (and Completely Made Up) Case of Quadruple Shah

Give me one rupee today, and
I'll give you two rupees
tomorrow and two rupees
day-after-tomorrow!

The Curious (and Completely Made Up) Case of Quadruple Shah



Give me one rupee today, and
I'll give you two rupees
tomorrow and two rupees
day-after-tomorrow!

This is clearly a **system**
with an input and
output!

The Curious (and Completely Made Up) Case of Quadruple Shah

Q. What happens if I give one rupee on Monday?



Give me one rupee today, and
I'll give you two rupees
tomorrow and two rupees
day-after-tomorrow!

The Curious (and Completely Made Up) Case of Quadruple Shah

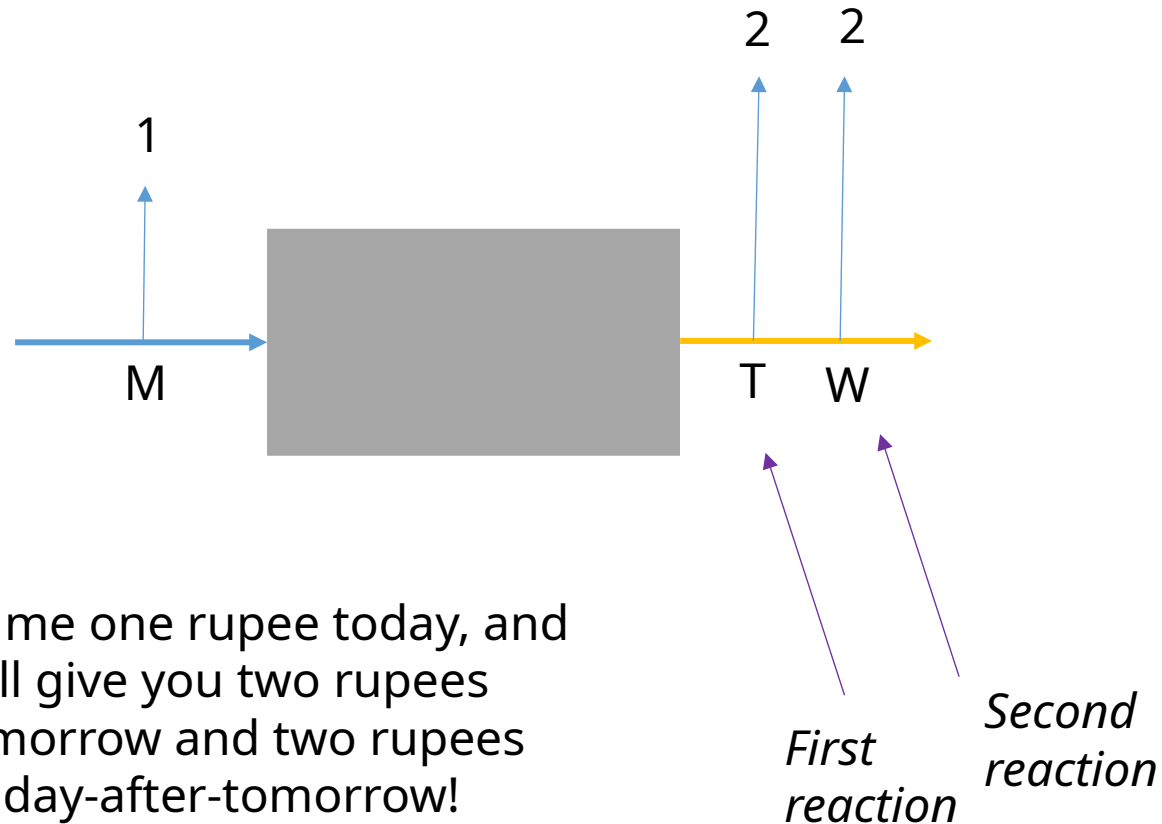
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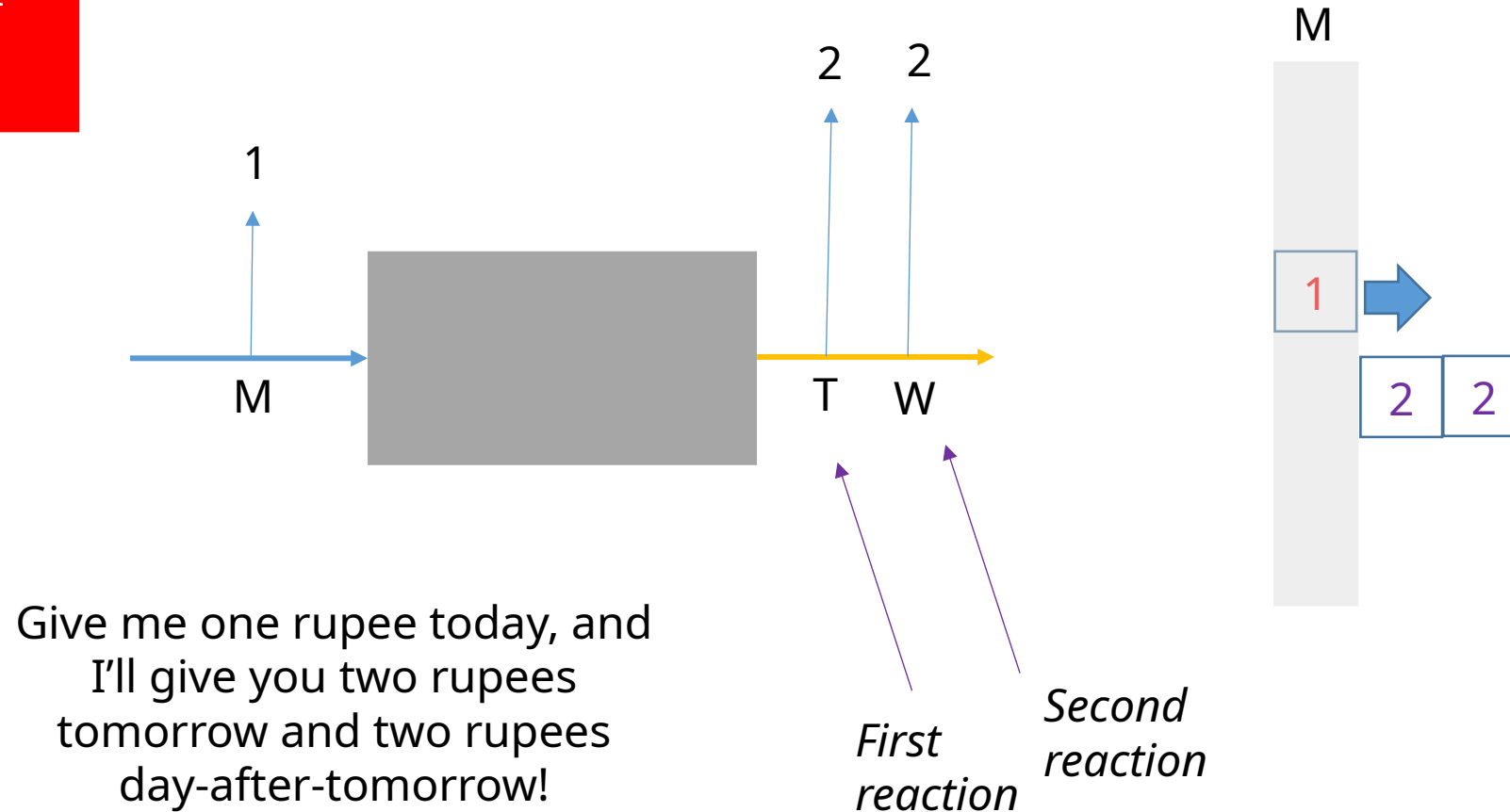
The Curious (and Completely Made Up) Case of Quadruple Shah

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The Curious (and Completely Made Up) Case of Quadruple Shah

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The Curious (and Completely Made Up) Case of Quadruple Shah

Q. What happens if I give one rupee on Monday and two on Tuesday?



The Curious (and Completely Made Up) Case of Quadruple Shah

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The Curious (and Completely Made Up) Case of Quadruple Shah

Q. What happens if I give one rupee on Monday and two on Tuesday?

The answer may be a bit **convoluted** this time!



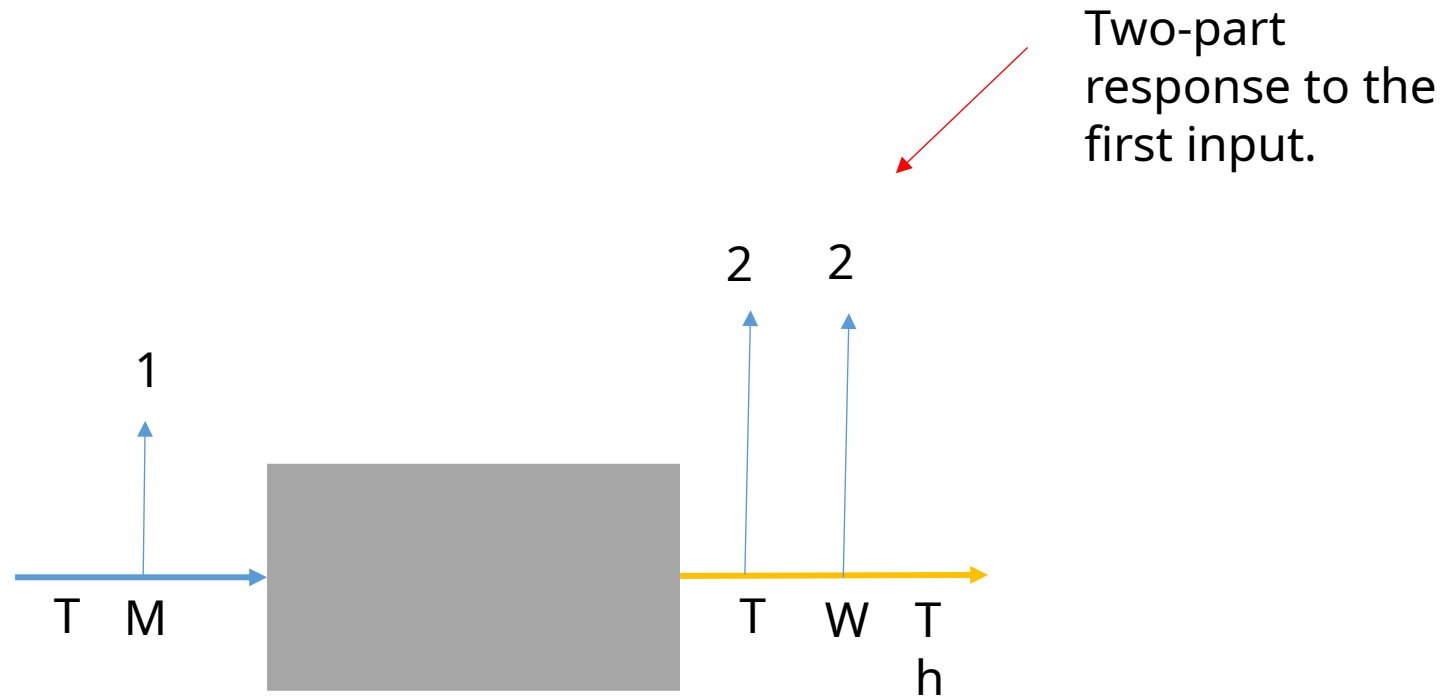
The Curious (and Completely Made Up) Case of Quadruple Shah

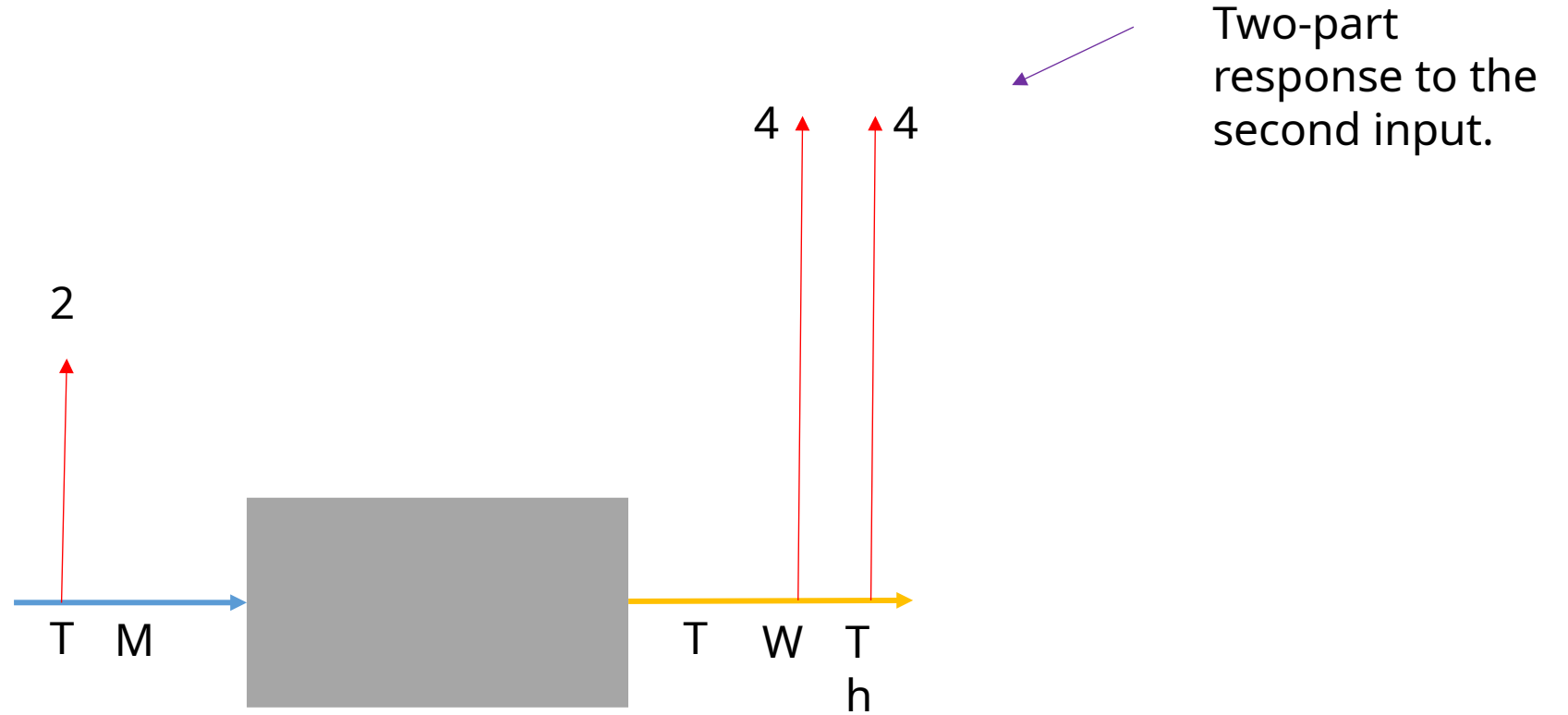
Q. What happens if I give one rupee on Monday and two on Tuesday?

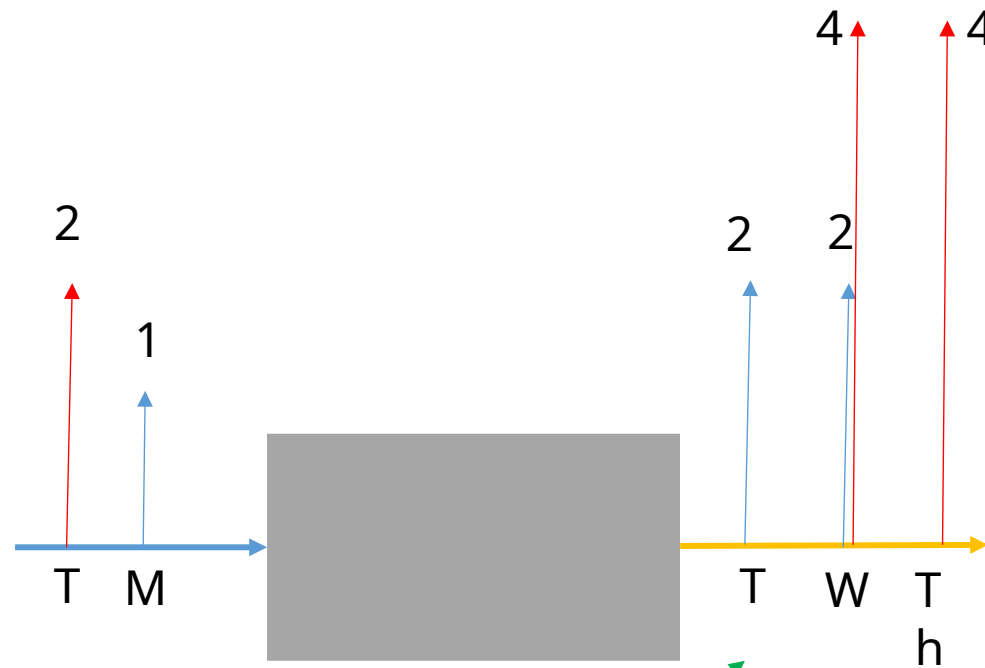
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Let's break it down...

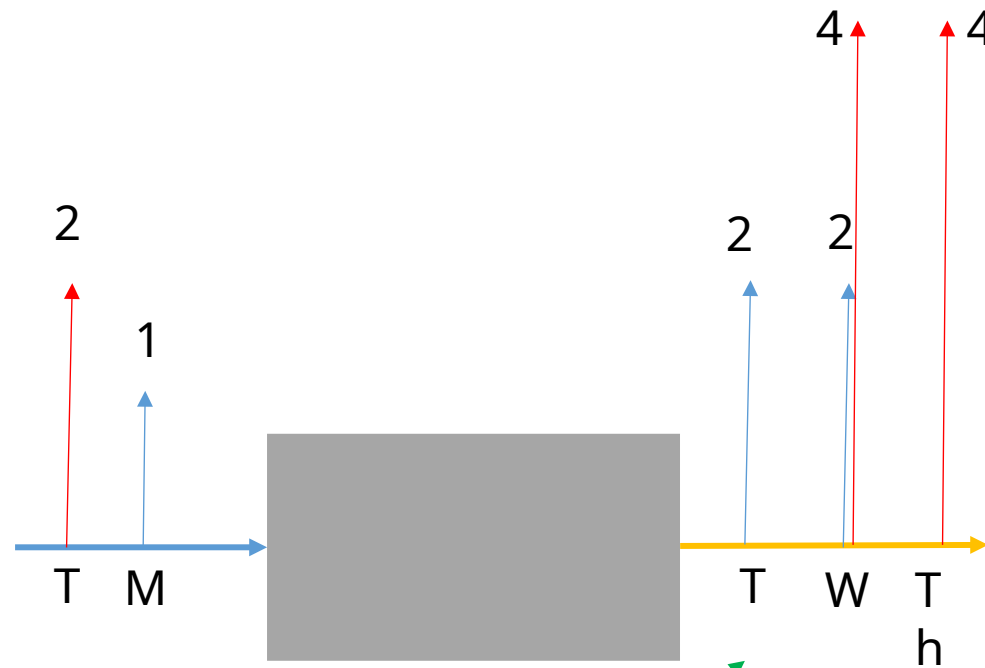






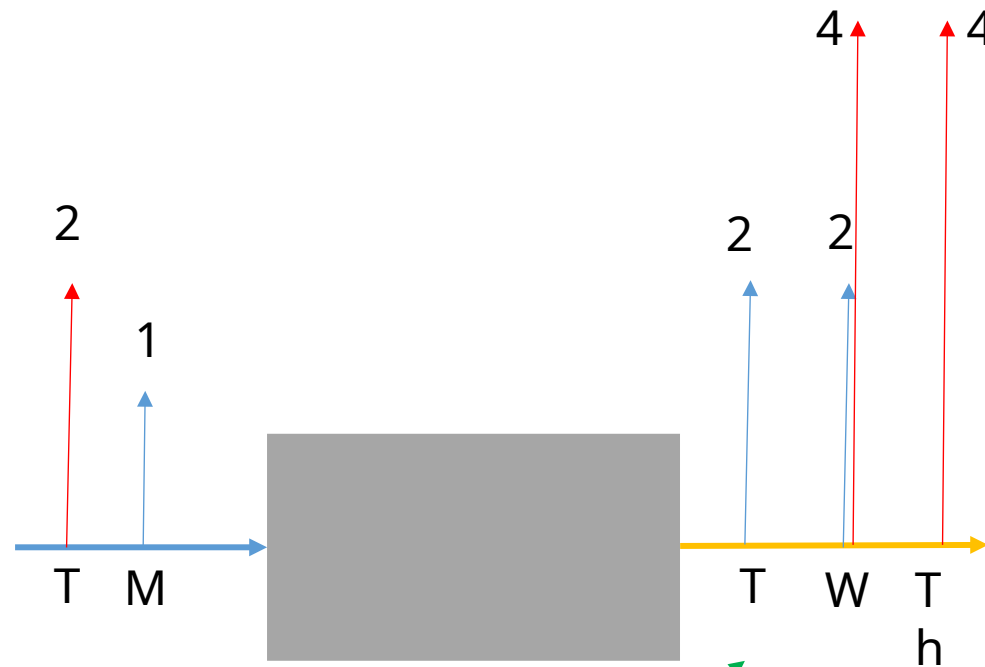


Joint (and bit
convoluted) response
to the two inputs



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to the two inputs

Tuesday	2
Wednesday	$2+4 = 6$
Thursday	4



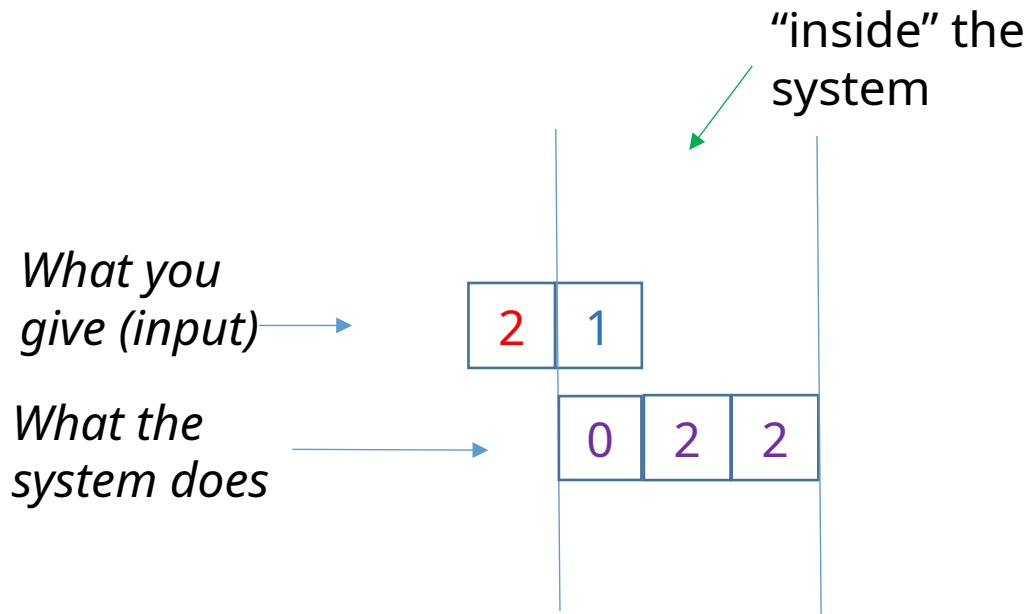
Tuesday	2
Wednesday	$2+4 = 6$
Thursday	4

Joint (and bit
convoluted) response
to the two inputs

Convolution helps us
calculate such outputs
mathematically.

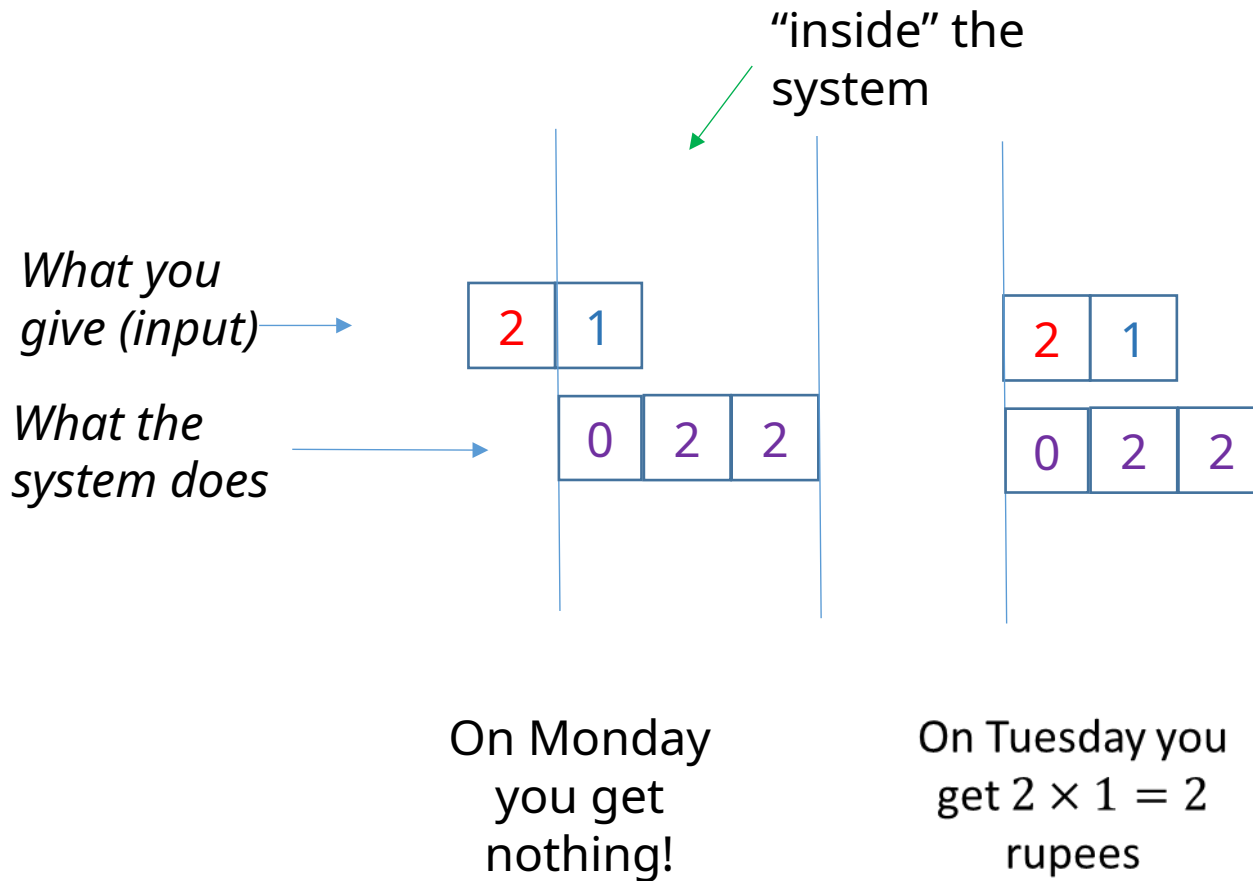
Visualizing Convolution: **Multiply overlapping cells and add the results! (then slide input right and repeat!)**

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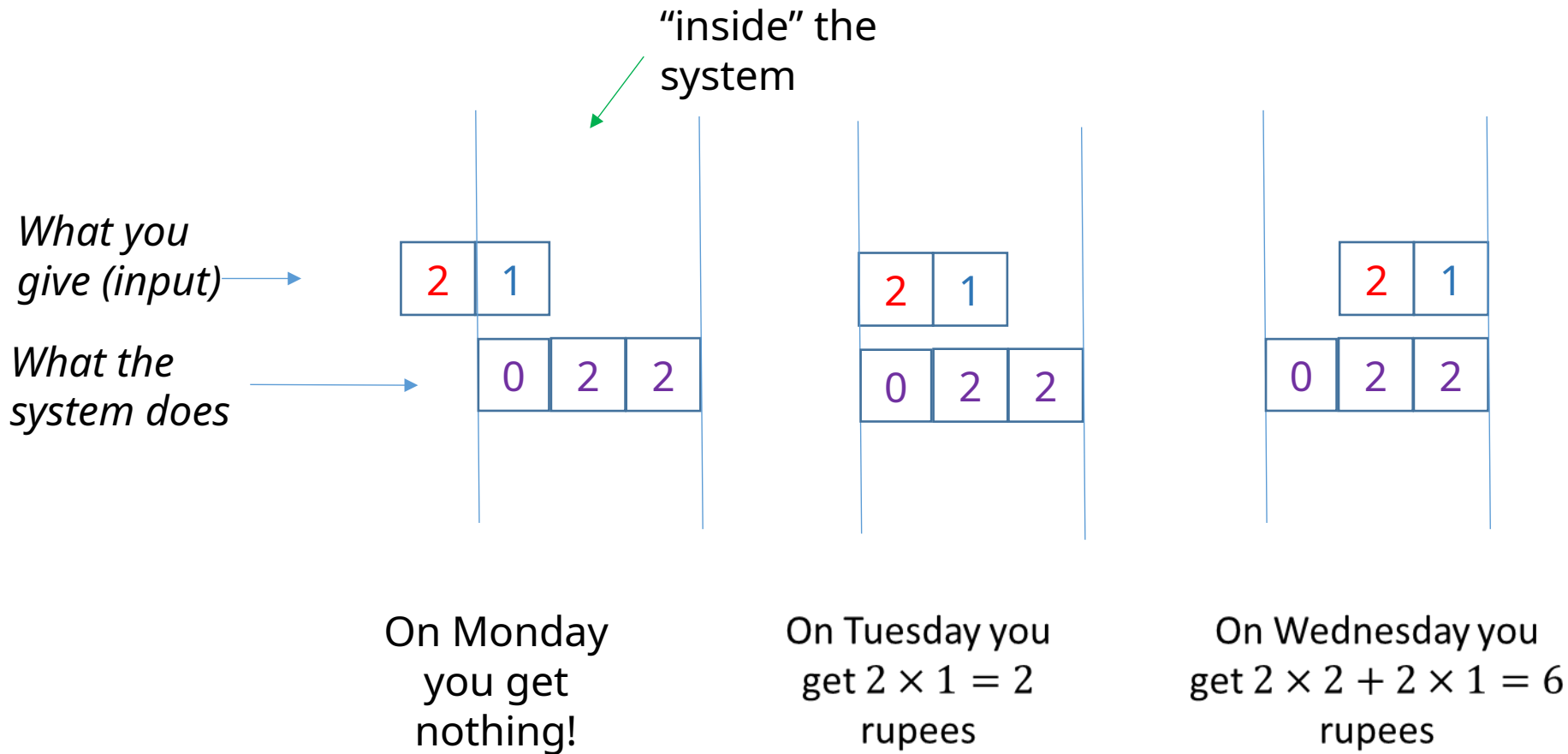


On Monday
you get
nothing!

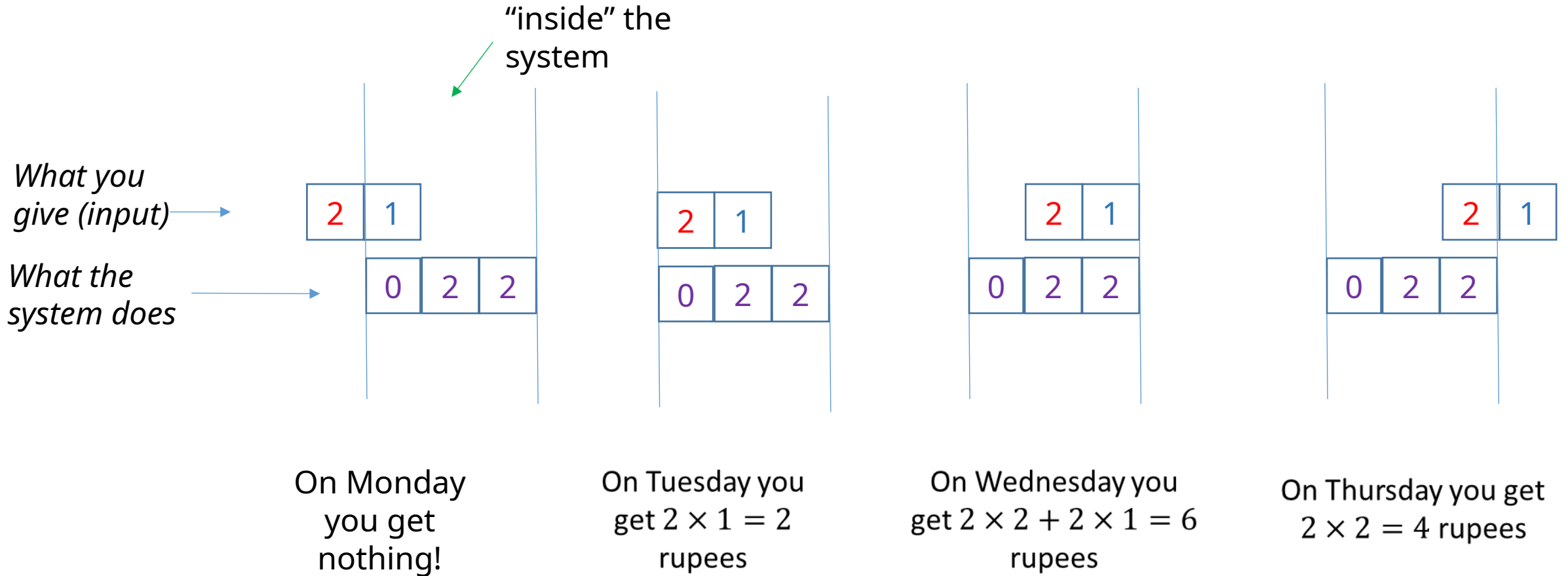
Visualizing Convolution: **Multiply overlapping cells and add the results! (then slide input right and repeat!)**



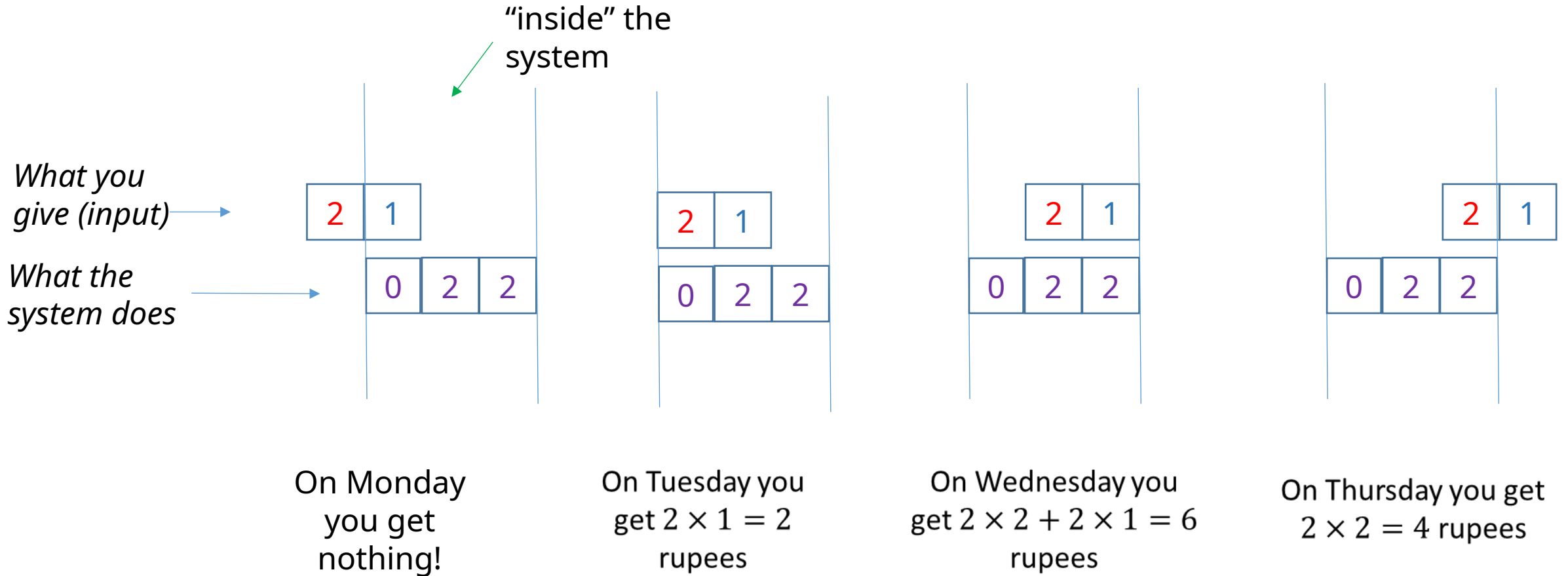
Visualizing Convolution: **Multiply overlapping cells and add the results! (then slide input right and repeat!)**



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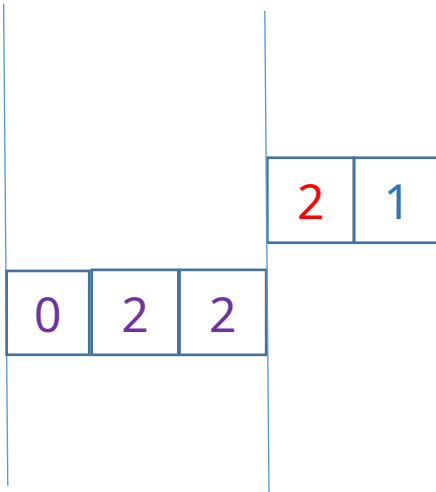


Visualizing Convolution: **Multiply overlapping cells and add the results! (then slide input right and repeat!)**



Q. What happens Friday onwards?

Visualizing Convolution: **Multiply overlapping cells and add the results! (then slide input right and repeat!)**



On Friday you
get nothing!

(an so on...)

Q. What happens Friday onwards?



All the notions introduced today will be covered in much greater detail in the course...

Questions?? Thoughts??

